MINING DEALER BEST PRACTICE

3500, 3400, C32, C27, C18 ENGINES PISTON PIN EXTRACTOR

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management

4.05-171128-02

1.0 Introduction

The extraction of the piston pin is a fundamental procedure in the engine disassembly process, since it secures the connecting rod with the piston and it is necessary to disassemble the piston regarding failure detection in the components.

To extract the piston pin from 3500, 3400, C32, C27, C18 engines, the technicians used to place the piston on a workbench or on a surface where they could turn the piston and insert the hand or any other tool to extract the piston pin exposing themselves to blows in the hands or unusual ergonomic positions as shown in figure 1. On the other hand, the bigger the piston, the more uncomfortable it will be to do the task.

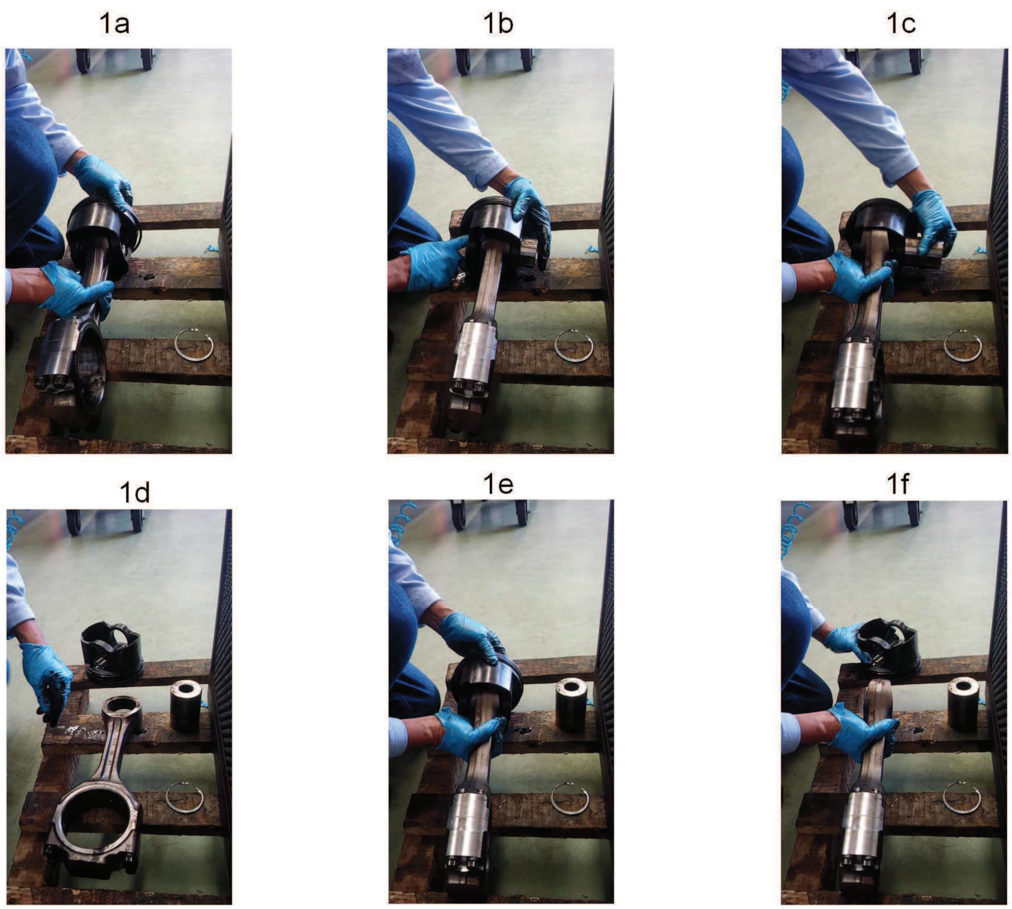


Figure1. Piston pin extraction before the improvement.

This best practice offers the design and manufacturing process of a tool to remove the piston pin when performing the piston disassembly procedure.

2.0 Best Practice Description

This BP applies to the piston pin of 3,500, 3,400, C32, C27, C18 engines. There was a feeling of discomfort when extracting the piston pin since there was no tool to extract the piston pin in an ergonomic way. The larger the piston, the more uncomfortable it is to do the job.

Geometrically, the tool has a T shape and it consists of a stem (12mm and 225mm in length) the lower end) is a coin-shaped bottom (30mm and 4mm thick).

The upper part has a widening of the cross section (20mm and 20mm in length on which there is a hole where a bar goes through. (13mm and a 100mm in length)

The aforementioned measurements gives the tool and overall dimension of 250mm in length and 100mm in width.

The tool is made of 4340 steel.

3.0 Implementation Steps

Design of the tool

Tool manufacturing process

Operation of the tool

- Insert the tool into the piston hole (3b)
- Once the tool is inserted; place it in such a way that the piston pin is above the circular edge of the lower part of the tool (3c)
- Remove the tool and the piston pin will come out as shown in the sequence of the figure (3d).

3a 3b 3c) 3d)

Figure 3. Piston pin removal process

4.0 Benefits

- It improves ergonomics and safety during the process.
- It reduces the time spent during the process.
- It reduces the risk of damage to parts involved in the process

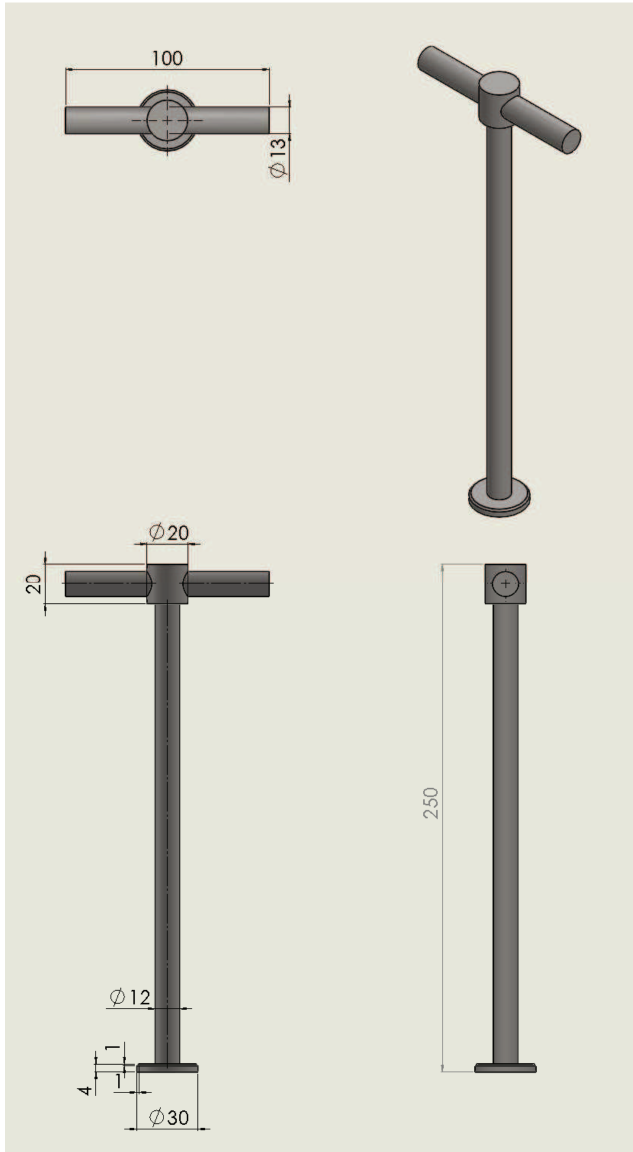
Model	Pin-Pistón P/N	Pin-Pistón P/N	Pin-Pistón P/N	Pin-Pistón P/N
3500	197-0560	138-8508	2638955	3023451
3400	8N-1608	173-0148		
C18	180-7350			
C27	180-7350			
C32	180-7350	524-5565		

Figure 4. Part number (Piston pin)

5.0 Resources Required

Manufacturing drawing and 1 welding technician
Approximate tool manufacturing cost $ 59.80 USD

6.0 Supporting Attachments / References



Tool Drawing

7.0 Related Best Practices

None

8.0 Acknowledgements

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